STSCI 5740 Project Report

Kelvin Chi (kc2346), Nina Wang (nw348), Tianle Pang (tp439), Xibei Xie (xx375)

1 Introduction

This project predicts 30-day hospital readmission for diabetes patients using the **Diabetes 130-US hospitals dataset (1999-2008)** from UCI. We compare regularized logistic regression, GAM, random forest, and XGBoost on the same cleaned patient-level dataset and evaluate mainly with ROC AUC due to class imbalance.

Project code: stsci5740-project

2 Data Cleaning

2.1 Patient-Level Deduplication

We retained only the first encounter per `patient_nbr` to avoid repeated-patient dependence. Sample size dropped from **101,766** to **71,518**.

2.2 Sparse Feature Removal

Removed high-missingness variables: - `weight` (96.01%) - `medical_specialty` (48.21%) - `payer_code` (43.41%)

2.3 Missing and Invalid Values

- `race`: missing -> `Other`
- `max_glu_serum`, `A1Cresult`: `None` kept as valid level
- `gender`: removed `Unknown/Invalid`
- `diag_1`: removed missing rows (11)
- `diag_2`, `diag_3`: missing -> `None`

2.4 Diagnosis Consolidation and Label Mapping

`diag_1`, `diag_2`, `diag_3` ICD-9 codes were grouped into 10 broader clinical categories to reduce high-cardinality levels. Administrative ID fields were mapped to descriptive categorical labels.

2.5 Target Variable and Final Dataset

Defined `readmitted_binary`: - `1`: original `readmitted = <30` - `0`: original `readmitted = >30` or `NO`

Final cleaned dataset: **71,504 observations**, **47 variables**.

3 Exploratory Data Analysis (EDA)

3.1 Outcome Distribution

The target is imbalanced: - `readmitted_binary = 0`: **65,213** - `readmitted_binary = 1`: **6,291**

Minority class rate is about **8.8%**, so ROC AUC is emphasized over threshold-based accuracy metrics.

3.2 Categorical Patterns

Key patterns from faceted count plots: - `race` is dominated by Caucasian, then African American - age is concentrated in older groups (especially **60-70**) - administrative and medication variables are often highly unbalanced - positive readmission remains the minority across most levels

3.3 Numerical Distributions

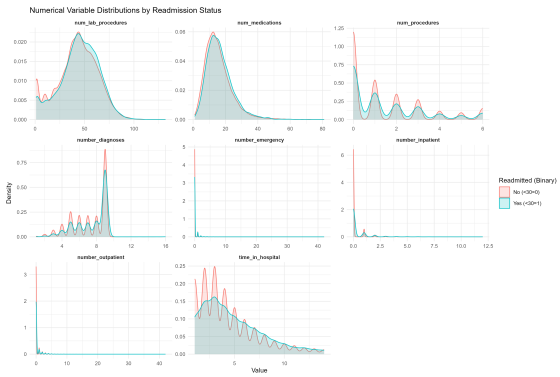

Figure 1: Numerical predictor density plots by readmission status.

Most utilization variables are right-skewed with long tails and substantial overlap between classes.

3.4 Correlation Structure

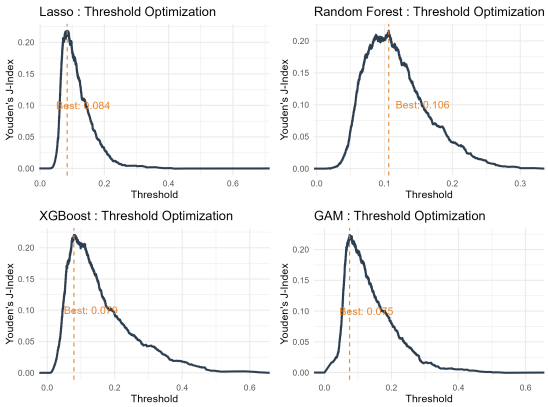

Figure 2: Correlation heatmap for numerical predictors.

Most pairwise correlations are low to moderate; no severe multicollinearity pattern is evident.

4 Linear Modeling

We used regularized logistic models as interpretable baselines in a high-dimensional encoded feature space: - Lasso (`L_1`) for sparsity - Ridge (`L_2`) for coefficient shrinkage stability

4.1 Preprocessing and Split

1. Stratified 80/20 train-test split by `readmitted_binary`.
2. One-hot encoding for categorical predictors.
3. Standardization for numeric predictors.
4. Same recipe/workflow applied to test data.

4.2 Regularized Logistic Regression

Elastic net was tuned by cross-validation on ROC AUC over: - `penalty` (`lambda`) - `mixture` (`alpha`, 0 = Ridge, 1 = Lasso)

4.2.1 Best Parameters

Penalty (`lambda`)	Mixture (`alpha`)	Selected Model	Config
0.0774	0	Ridge Logistic Regression	`pre0_mod17_post0`

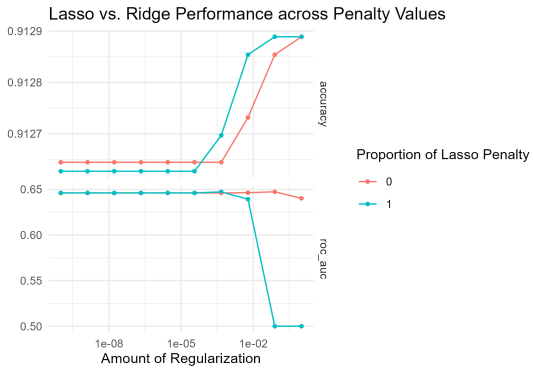

Figure 3: Cross-validation tuning surface for logistic penalty search.

Ridge slightly outperformed Lasso, suggesting signal is distributed across many predictors.

4.2.2 Important Predictors

Predictor	Coefficient	Interpretation
`number_inpatient`	+0.1360	Strongest positive readmission signal
Discharged / transferred to another rehab facility	+0.1200	Higher post-discharge risk
Discharged to home	-0.0889	Lower relative risk
Expired	-0.0825	Negative due to coding/terminal outcome
Discharged to SNF	+0.0675	Higher risk profile
`time_in_hospital`	+0.0425	Longer stay, higher severity
`number_diagnoses`	+0.0365	Greater comorbidity burden
`diag_1_Respiratory`	-0.0361	Lower relative risk vs baseline
Age 50-60	-0.0345	Slightly lower relative risk
`diabetesMed_Yes`	+0.0332	Medication may proxy severity

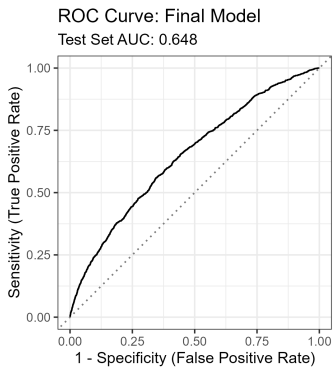

Figure 4: Test ROC curve for the final ridge logistic regression model.

Final Ridge test ROC AUC: **0.648**.

4.3 Generalized Additive Model (GAM)

GAM was fit on the same split (binomial family, logit link) to allow smooth nonlinear effects while keeping additive interpretability.

Smoothed Numerical Predictor	EDF	Linear/Categorical Predictor
`time_in_hospital`	3.30	`race`
`num_lab_procedures`	2.18	`gender`
`num_medications`	1.04	`age`
`number_diagnoses`	2.39	`diabetesMed`
`number_inpatient`	2.65	`insulin`
`number_emergency`	2.10	`discharge_disposition_id`

All smooth terms used `k = 5`; total EDF = **39.65**; UBRE = **-0.4338**.

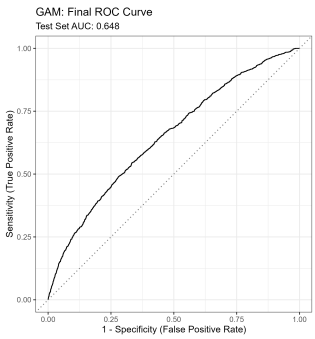

Figure 5: Estimated smooth effects from the fitted GAM model.

Test ROC AUC: **0.648**. Strongest nonlinear signal came from `number_inpatient` and `number_emergency`.

5 Nonlinear Modeling

5.1 Random Forest (RF)

RF was tuned by cross-validation over `mtry` and `min_n` (500 trees, ROC AUC objective).

- ROC AUC: **0.651**
- Accuracy: **0.909**
- Precision: **0.909**
- Recall: **1.000**
- F1: **0.952**

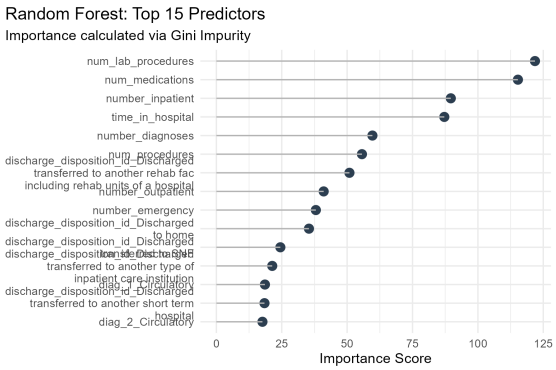

Figure 6: Random Forest variable importance ranked by Gini importance.

5.2 Gradient Boosted Trees (XGBoost)

XGBoost was tuned with racing over `tree_depth`, `min_n`, `loss_reduction`, `learn_rate`, and `sample_size` (1000 trees, early stopping = 10, ROC AUC objective).

- ROC AUC: **0.6558**
- Precision: **0.909**
- Recall: **1.000**
- F1: **0.952**

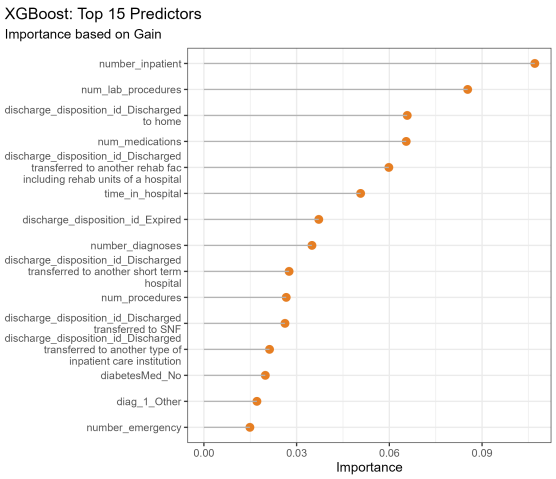

Figure 7: XGBoost variable importance ranked by gain.

XGBoost delivered the best discrimination among tested models.

6 Final Model Comparison

6.1 Test Set Performance

Model	ROC AUC	Precision	Recall	F1 Score
Ridge Logistic Regression	0.648	0.909	1.000	0.952
GAM	0.648	0.909	1.000	0.952
Random Forest	0.651	0.909	1.000	0.952
XGBoost	**0.656**	0.909	1.000	0.952

All models have similar threshold-based metrics due to class imbalance and default cutoff behavior. ROC AUC is the key differentiator.

6.2 Default-Threshold Behavior

Metric	Ridge Logistic Regression	GAM	Random Forest	XGBoost
Predicted positive cases	2	6	0	10

7 Threshold Optimization

Default cutoff (`0.50`) produced majority-class predictions. We re-optimized each model threshold using Youden's J statistic:

[J = Sensitivity + Specificity - 1]

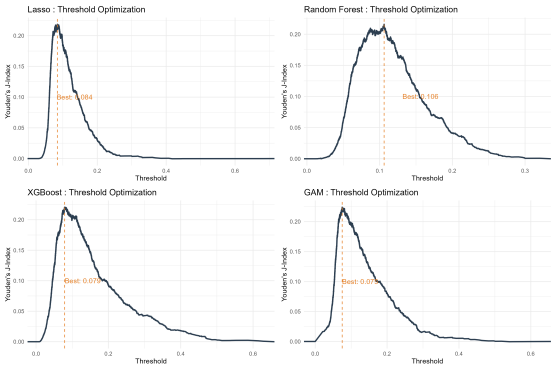

Figure 8: Youden's J-statistic curves used to select optimized probability thresholds.

7.1 Tuned Confusion Matrices

Ridge (t = 0.084)

Pred\True	N	Y
N	7,928	501
Y	5,090	782

RF (t = 0.106)

Pred\True	N	Y
N	10,027	727
Y	2,966	581

XGB (t = 0.079)

Pred\True	N	Y
N	7,653	482
Y	5,340	826

GAM (t = 0.075)

Pred\True	N	Y
N	7,298	432
Y	5,720	851

Threshold tuning increased sensitivity for minority-class readmissions, with more false positives.

- Most true positives: **GAM (851)**
- Next: **XGBoost (826)**
- Ridge: **782**
- Most conservative after tuning: **RF (581)**

8 Final Conclusion

All models outperformed random guessing, with ROC AUC from **0.648 to 0.656**. **XGBoost** had the best overall discrimination (**0.656**), with **Random Forest** second (**0.651**). **Ridge** and **GAM** remained competitive (**0.648** each).

Important predictors were consistent across models: prior utilization, treatment intensity, disease burden, and discharge disposition.

Given class imbalance, threshold selection strongly affects operational usefulness. After tuning, **GAM and XGBoost** gave the strongest readmission detection.